

Math 135

Please give me feedback! I am at the first row.

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§1 Admin

§1.1 General Information

§1.1.1 Contact

For mathematics questions: abeltaos@uwaterloo.ca

For course logistics: math135@uwaterloo.ca

§1.1.2 Teaching Style

The LEARN website contains several useful resources. In particular:

- (i) The icons at the top of the page provide quick access to important resources, including the course schedule.
- (ii) Crowdmark is used for submitting assignments.

§1.1.3 Graded Assignments

There are three assignments each week.

§1.1.4 Late Submission Policy

Late submissions are accepted with a penalty of 1% per minute for the entire assignment.

§1.1.5 Office Hours

Monday and Wednesday 2:00 to 3:30 in MC 6503 or by appointment (email the instructor to arrange).

§2 Introduction to the Language of Mathematics

§2.1 Introduction to Proof

Definition 2.1.1 (Proof)

A proof is a logically valid argument that establishes the truth of a mathematical statement using definitions, axioms, and previously proved results.

Proposition 2.1.2

For every positive integer n , the number $n^2 + 1$ is not a perfect square.

Proof

Let n be a positive integer. The consecutive perfect squares surrounding $n^2 + 1$ are n^2 and $(n + 1)^2$. Indeed,

$$n^2 < n^2 + 1 < n^2 + 2n + 1 = (n + 1)^2,$$

where the second inequality follows from $n > 0$.

Therefore, $n^2 + 1$ lies strictly between two consecutive perfect squares, so it cannot itself be a perfect square.

Q.E.D.

§2.2 Sets

Definition 2.2.1 (Set)

A **set** is a well-defined, unordered collection of distinct objects. Each object that appears in this collection is called an **element** of the set.

For a finite set, we can denote it by listing its elements, separated by commas, surrounding the entire list by braces $\{ \}$.

Example 2.2.2

The set containing the elements 2, 7, and -3 can be denoted by

$$\{7, -3, 2\}.$$

The order of listing is irrelevant, so $\{2, 7, -3\}$ is the same set.

Example 2.2.3

Here are some examples:

- $\{2, 4, 6, 8\}$
- $\{1, 2, \{1, 2, 3\}\}$
- $\{a, b, c, d\}$
- $\mathbb{N} = \{1, 2, 3, \dots\}$

- $\mathbb{Z} = \{\dots, -1, 0, 1, \dots\}$
- $\mathbb{Q} = \{\frac{a}{b} : a, b \in \mathbb{Z}, b \neq 0\}$

An element may itself be a set: $\{1, 2, \{1, 2, 3\}\}$ has exactly three elements, namely 1, 2, and $\{1, 2, 3\}$.

Remark

Infinite sets can have different sizes: $|\mathbb{N}| = \aleph_0$ while $|\mathbb{R}| = 2^{\aleph_0}$, and $2^{\aleph_0} > \aleph_0$. Whether $2^{\aleph_0}$ equals the next size $\aleph_1$ is the *continuum hypothesis*, which can be neither proved nor disproved from the standard axioms, so $|\mathbb{R}| = \aleph_1$ should not be written as a fact.

Definition 2.2.4

The **parity** of an integer is **even** if the integer is a multiple of 2, and **odd** otherwise.

§2.3 Statement and Negation

Definition 2.3.1 (Statement)

A **statement** is a sentence that has a definite state of being either true or false.

Definition 2.3.2 (Negation)

Suppose that A is a statement. Then the negation of A , denoted by $\neg A$, is the statement asserting the opposite truth value to A . That is, $\neg A$ is false when A is true, and $\neg A$ is true when A is false.

§2.4 Quantifiers and Quantified Statements

§2.4.1 Universal Quantifier

$\forall$ means “for all”, and it is called the universal quantifier. The statement $\forall x \in S, P(x)$ is true when $P(x)$ is true for *every* element x of S .

Example 2.4.1

“Every integer $n \geq 5$ satisfies $2^n > n^2$ ” can be written as

$$\forall n \in \{n \in \mathbb{Z} : n \geq 5\}, 2^n > n^2.$$

Checking $n = 5, 6, 7$ is not a proof: a universal statement must hold for every n in the domain. (The domain matters: the statement fails at $n = 3$, since $8 < 9$.)

The negation of a universally quantified statement is

$$\neg (\forall x \in S, P(x)) \equiv (\exists x \in S, \neg P(x)).$$

So a single counterexample is enough to show that a universal statement is false.

§2.4.2 Existential Quantifier

$\exists$ means “there exists”, and it is called the existential quantifier. The statement $\exists x \in S, P(x)$ is true when $P(x)$ is true for *at least one* $x \in S$.

Example 2.4.2

$$\exists a \in \mathbb{Z}, a^2 + 29a + 209 \leq 0.$$

This is true: one witness is enough, and $a = -14$ gives $196 - 406 + 209 = -1 \leq 0$.

The negation of an existentially quantified statement (such as $\exists x \in S, P(x)$) is

$$\neg (\exists x \in S, P(x)) \equiv (\forall x \in S, \neg P(x)).$$

Remark

A quantified statement contains four parts:

- variable;
- quantifier;
- domain;
- an open sentence involving the variable (that is either true or false whenever a value of the variable is chosen from the domain).

§2.4.3 Nested Quantifiers

Consider the statement

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, x > y.$$

This statement is true: given any $x \in \mathbb{R}$, we may choose $y = x - 1$, and then $x > y$. Note that y is allowed to depend on x .

Remark

Order matters! Consider the statement $\exists y \in \mathbb{R}, \forall x \in \mathbb{R}, x > y$. It says there is a single real number y which is smaller than every real number, including y itself. This is false: taking $x = y$ gives $y > y$.

To negate a nested quantified statement, negate one quantifier at a time from the outside in: each $\forall$ becomes $\exists$, each $\exists$ becomes $\forall$, and the open sentence is negated. For example,

$$\begin{aligned} \neg (\forall x \in S, \exists y \in T, P(x, y)) &\equiv \\ \exists x \in S, \forall y \in T, \neg P(x, y). \end{aligned}$$

§2.5 Summation and Product Notation

Writing $1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10$ is manageable, but writing out every term through 1000 is not. Summation notation specifies both the terms and their range, instead of relying on dots in $1 + 2 + \cdots + 1000$.

Definition 2.5.1 (Summation notation)

For integers $a \leq b$ and terms $t_a, \dots, t_b$, write

$$\sum_{i=a}^b t_i = t_a + t_{a+1} + \dots + t_b.$$

The index i runs through every integer from a to b .

Example 2.5.2

$$1 + 2 + \dots + 100 = \sum_{i=1}^{100} i, \quad 1 + 8 + 27 + 64 + 125 = \sum_{i=1}^5 i^3.$$

In the second sum, the term indexed by i is i^3 .

§2.5.1 Properties of Summations

For terms x_i, y_i , a constant c , and integer bounds $a \leq b$ and $m \leq n$, the following identities let us rewrite sums without changing their terms:

(i) Multiplying by a constant:

$$c \sum_{i=a}^b x_i = \sum_{i=a}^b cx_i.$$

(ii) Adding or subtracting sums with the same bounds:

$$\sum_{i=a}^b x_i \pm \sum_{i=a}^b y_i = \sum_{i=a}^b (x_i \pm y_i).$$

(iii) Shifting the index by an integer r :

$$\sum_{i=m}^n x_i = \sum_{j=m+r}^{n+r} x_{j-r}.$$

The new index $j = i + r$ names the same terms.

(iv) Splitting a sum at n , where $2 \leq n \leq k$:

$$\sum_{i=1}^{n-1} i + \sum_{i=n}^k i = \sum_{i=1}^k i.$$

§2.5.2 Product Notation

Products use the same indexing convention, with multiplication in place of addition. For $n \geq 1$,

$$\prod_{i=1}^n i = 1 \cdot 2 \cdot 3 \cdots n = n!.$$

Zero sum/product notation

We have

$$\sum_{i=m}^m x_i = x_m \quad \text{and} \quad \prod_{i=m}^m x_i = x_m.$$

Empty products / sums

$$\sum_{j=5}^4 = 0 \quad \text{and} \quad \prod_{i=10}^5 i^2 = 1.$$

§3 Logical Analysis of Mathematical Statements

§3.1 Truth Table and Negation

Definition 3.1.1

The truth value for “not A ”, written symbolically as $\neg A$, is defined by the truth table given below.

A	$\neg A$
T	F
F	T

We also refer to $\neg A$ as the negation of A .

Here A is a **statement variable**: it stands for an arbitrary statement, so a truth table must list every possible truth value of A .

Definition 3.1.2 (Logical equivalence)

Two compound statements are **logically equivalent**, written $\equiv$, if they have the same truth value in every row of their truth table, that is, for every assignment of truth values to their statement variables.

So a truth table can prove that two statements are logically equivalent.

Example 3.1.3

Given the truth table for A , $\neg A$, $\neg(\neg A)$,

A	$\neg A$	$\neg(\neg A)$
T	F	T
F	T	F

The first and last columns agree in every row, so this truth table serves as a proof that $A \equiv \neg(\neg A)$.

§3.2 Conjunction and Disjunction

Definition 3.2.1 (Compound statement)

A compound statement is a statement composed of several individual statements, each of which is called a component statement.

Definition 3.2.2

The truth value for $A \wedge B$, is defined by the truth table

A	B	$A \wedge B$
T	T	T
T	F	F
F	T	F
F	F	F

We also refer to $A \wedge B$ as the **conjunction** of A and B .

Each statement variable has two possible truth values, so with three statement variables we need $2^3 = 8$ rows to cover all the possibilities (and 2^n rows for n variables).

Example 3.2.3

The compound statement

$$(2 + 3 = 6) \wedge (\pi + 2 > 5)$$

is false, since the statement $2 + 3 = 6$ is false.

Definition 3.2.4

The truth value for A or B , written as $A \vee B$, is defined by the truth table

A	B	$A \vee B$
T	T	T
T	F	T
F	T	T
F	F	F

We also refer to $A \vee B$ as the **disjunction** of A and B .

Remark

The disjunction $A \vee B$ is the *inclusive* or: it is true whenever at least one of A , B is true, including the case where both are true. This differs from the *exclusive* or (xor), written $A \oplus B$, which is true when exactly one of A , B is true and false when both are true. Thus $A \vee B$ and $A \oplus B$ agree in every case except $A = B = \text{T}$, where $A \vee B$ is true but $A \oplus B$ is false.

Example 3.2.5

The compound statement

$$(2 + 3 = 6) \vee (\pi + 2 \geq 5)$$

is true, since its second component $\pi + 2 \geq 5$ is true.

Example 3.2.6

Let us use a truth table to prove that

$$\neg(A \vee B) \equiv (\neg A \wedge \neg B)$$

A	B	$A \vee B$	$\neg(A \vee B)$	$\neg A$	$\neg B$	$\neg A \wedge \neg B$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

The columns for $\neg(A \vee B)$ and $\neg A \wedge \neg B$ agree in all four rows.

A similar equivalence is

$$\neg(A \wedge B) \equiv ((\neg A) \vee (\neg B)).$$

Together, these two equivalences are called De Morgan's Laws: negating swaps $\wedge$ and $\vee$.

Theorem 3.2.7 (De Morgan's Laws)

For statement variables A and B , we have

$$\neg(A \vee B) \equiv (\neg A \wedge \neg B)$$

$$\neg(A \wedge B) \equiv (\neg A \vee \neg B)$$

Proposition 3.2.8

For statements A, B, C , the following rules hold for the logical operations $\wedge$ and $\vee$:

Commutative Laws

$$A \vee B \equiv B \vee A$$

$$A \wedge B \equiv B \wedge A$$

Associative Laws

$$A \vee (B \vee C) \equiv (A \vee B) \vee C$$

$$A \wedge (B \wedge C) \equiv (A \wedge B) \wedge C$$

Distributive Laws

$$A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C)$$

$$A \wedge (B \vee C) \equiv (A \wedge B) \vee (A \wedge C)$$

§3.3 Implication

Definition 3.3.1

The truth value for “ A implies B ”, written symbolically as $A \implies B$, is defined by the truth table

A	B	$A \implies B$
T	T	T
T	F	F
F	T	T
F	F	T

We also refer to $A \implies B$ as an implication.

In $A \implies B$, the statement A is the **hypothesis** (what we assume to be true) and B is the **conclusion** (what we must then show). The only way for an implication to be false is a true hypothesis with a false conclusion. When A is false, $A \implies B$ is true no matter what B is (it is *vacuously* true).

Example 3.3.2

$$(\forall x \in \mathbb{R}) \left[(x > 2) \implies (x^2 > 4) \right]$$

This is true. If $x > 2$, then $x > 0$, so $x^2 = x \cdot x > 2x > 4$. If $x \leq 2$, the hypothesis is false, so the implication is vacuously true for that x .

Remark

An implication with an “or” in its hypothesis splits into two implications:

$$(A \vee B) \implies C \equiv (A \implies C) \wedge (B \implies C).$$

This is the logic behind a proof by cases.

§3.4 Converse and Contrapositive

§3.4.1 Converse

The implication $B \implies A$ is called the **converse** of $A \implies B$.

A	B	$A \implies B$	$B \implies A$
T	T	T	T
T	F	F	T
F	T	T	F
F	F	T	T

Remark

$(A \implies B) \not\equiv (B \implies A)$: the two columns differ in the second and third rows. For instance, $\forall x \in \mathbb{R}, (x > 2) \implies (x^2 > 4)$ is true, but its converse $\forall x \in \mathbb{R}, (x^2 > 4) \implies (x > 2)$ is false (take $x = -3$).

§3.4.2 Contrapositive

The implication $(\neg B) \implies (\neg A)$ is called the **contrapositive** of $A \implies B$.

A	B	$\neg A$	$\neg B$	$A \implies B$	$(\neg B) \implies (\neg A)$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Remark

$(A \implies B) \equiv ((\neg B) \implies (\neg A))$: an implication and its contrapositive always have the same truth value.

An implication is also logically equivalent to a disjunction, as shown in the following truth table:

A	B	$\neg A$	$A \implies B$	$\neg A \vee B$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

That is, $A \implies B \equiv \neg A \vee B$.

Remark

The negation of an implication is not an implication. Negating $\neg A \vee B$ with De Morgan's Law gives

$$\neg(A \implies B) \equiv A \wedge \neg B,$$

which matches the only row where $A \implies B$ is false.

§3.4.3 If and Only If**Definition 3.4.1**

The truth value for “ A if and only if B ”, written symbolically as $A \iff B$, is defined by the truth table

A	B	$A \iff B$
T	T	T
T	F	F
F	T	F
F	F	T

Thus $A \iff B$ is true exactly when A and B have the same truth value, and

$$(A \iff B) \equiv (A \implies B) \wedge (B \implies A).$$

To prove an “if and only if” statement, we prove an implication and its converse.

§4 Methods of Mathematical Proof

§4.1 Proving Universally Quantified Statements

A **proposition** is a true statement that has been proved to be true. A **theorem** is a particularly significant proposition. A **lemma** is a subsidiary proposition. A **corollary** is a proposition that follows almost immediately from a theorem.

Example 4.1.1

The law of cosines is a theorem: in a triangle with sides a, b, c , where θ is the angle between a and b ,

$$c^2 = a^2 + b^2 - 2ab \cos \theta.$$

The Pythagorean theorem is a corollary: taking $\theta = \pi/2$ gives $\cos \theta = 0$, so $c^2 = a^2 + b^2$.

To prove $\forall x \in S, P(x)$, let x be an *arbitrary* element of S , using only the fact that $x \in S$, and show that $P(x)$ is true. Since nothing special about x was assumed, the argument works for every element of S .

Let's do our first proof.

Proposition 4.1.2

$$\forall x \in \mathbb{R}, x^2 + 1 \geq 2x.$$

Remark (Proof idea)

Moving everything to one side gives $x^2 - 2x + 1 \geq 0$, and the left side is the perfect square $(x - 1)^2$. Since squares of real numbers are never negative, we start from $(x - 1)^2 \geq 0$ and work forward.

Proof

Let x be an arbitrary real number. Therefore, $x - 1$ must also be a real number, and hence

$$(x - 1)^2 \geq 0.$$

Expanding the left side, we get

$$x^2 - 2x + 1 \geq 0.$$

Adding $2x$ to both sides yields

$$x^2 + 1 \geq 2x.$$

Q.E.D.

Proposition 4.1.3

$$\forall x, y \in \mathbb{R}, x^4 + x^2y + y^2 \geq 5x^2y - 3y^2.$$

Remark (Proof idea)

Moving everything to the left side gives $x^4 - 4x^2y + 4y^2 \geq 0$, and $x^4 - 4x^2y + 4y^2 = (x^2 - 2y)^2$. This explains the choice of $x^2 - 2y$ below.

Proof

Let x and y be arbitrary real numbers. Then $x^2 - 2y$ is also a real number. So

$$(x^2 - 2y)^2 \geq 0.$$

Expanding, we get

$$x^4 - 4x^2y + 4y^2 \geq 0.$$

Adding $5x^2y$ to both sides, we have

$$x^4 + x^2y + 4y^2 \geq 5x^2y.$$

Now, subtract $3y^2$ from both sides:

$$x^4 + x^2y + y^2 \geq 5x^2y - 3y^2.$$

Q.E.D.

Proposition 4.1.4

$$\forall x, y \in \mathbb{R}, \max(x, y) = \frac{x + y + |x - y|}{2}.$$

Remark (Proof idea)

Both $\max(x, y)$ and $|x - y|$ are defined by cases depending on which of x, y is larger, so we split into the cases $x \geq y$ and $x < y$. These cases cover every pair of real numbers.

Proof

Let x and y be arbitrary real numbers.

Case 1: If $x \geq y$, then $\max(x, y) = x$, and $x - y \geq 0$, so $|x - y| = x - y$.

Then

$$\frac{x + y + |x - y|}{2} = \frac{x + y + x - y}{2} = \frac{2x}{2} = x = \max(x, y).$$

So the claim holds when $x \geq y$.

Case 2: If $x < y$, then $\max(x, y) = y$, and $x - y < 0$. So

$$|x - y| = -(x - y) = y - x.$$

Then

$$\frac{x + y + |x - y|}{2} = \frac{x + y + y - x}{2} = \frac{2y}{2} = y = \max(x, y).$$

The claim holds in both cases, so it holds for all $x, y \in \mathbb{R}$.

Q.E.D.

§4.1.1 Disproving a Universal Statement

Since $\neg(\forall x \in S, P(x)) \equiv \exists x \in S, \neg P(x)$, to disprove a universal statement it suffices to give one **counterexample**: a specific $x \in S$ for which $P(x)$ is false.

Example 4.1.5

Disprove:

$$\forall x \in \mathbb{R}, (x^2 - 1)^2 > 0.$$

Solution. Take $x = 1$. Then

$$(x^2 - 1)^2 = (1 - 1)^2 = 0 \not> 0.$$

So $x = 1$ is a counterexample, and the statement is false.

Exercises

- Prove $\forall \theta \in \mathbb{R}, \sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$.
- Prove $\forall x \in \mathbb{R}, |x - 3| + 2|x + 2| \geq 5$.

§4.2 Proof by Contrapositive, Elimination

Several examples below use divisibility.

Definition 4.2.1 (Divisibility)

Let $a, b \in \mathbb{Z}$. We say a **divides** b , written $a \mid b$, if

$$\exists k \in \mathbb{Z}, b = ak.$$

Otherwise we write $a \nmid b$.

Definition 4.2.2 (Proof by Contrapositive)

Since $(A \implies B) \equiv (\neg B \implies \neg A)$, we can prove $(A \implies B)$ by proving that $((\neg B) \implies (\neg A))$. This method is useful when a direct proof of $A \implies B$ is awkward, but assuming $\neg B$ gives a concrete hypothesis that is easier to work with.

Example 4.2.3

Here is an example of contrapositive proof.

Proposition 4.2.4

For every integer x , if $4 \nmid x$ then $36 \nmid 3x$.

Proof

The hypothesis $4 \nmid x$ says only what x is *not*, which is hard to use directly, so we prove the contrapositive

$$(36 \mid 3x) \implies (4 \mid x).$$

Let x be an arbitrary integer. Assume that $36 \mid 3x$. Thus, there exists an integer k such that $3x = 36k$.

Dividing both sides by 3,

$$x = 12k = 4 \cdot (3k) \text{ and } 3k \in \mathbb{Z},$$

so $4 \mid x$.

Therefore, by the contrapositive method,

$$(\forall x \in \mathbb{Z}) [4 \nmid x \implies 36 \nmid 3x].$$

Q.E.D.

Example 4.2.5

Here is another example.

Proposition 4.2.6

$$(\forall a, b \in \mathbb{R}) [(ab \in \mathbb{R} \setminus \mathbb{Q}) \implies (a \in \mathbb{R} \setminus \mathbb{Q} \vee b \in \mathbb{R} \setminus \mathbb{Q})]$$

Proof

Irrationality is a “not” condition, while rationality gives us fractions to compute with. By De Morgan’s Law, the negation of the conclusion is “ $a \in \mathbb{Q}$ and $b \in \mathbb{Q}$ ”, so the contrapositive is: $\forall a, b \in \mathbb{R}$, if $a \in \mathbb{Q}$ and $b \in \mathbb{Q}$, then $ab \in \mathbb{Q}$.

Let a, b be arbitrary in $\mathbb{R}$. Assume that a, b are both rational.

Then there exist $p, q, r, s \in \mathbb{Z}$ with $q \neq 0$ and $s \neq 0$ such that

$$a = \frac{p}{q} \quad \text{and} \quad b = \frac{r}{s}.$$

Thus,

$$a \cdot b = \frac{pr}{qs} \quad \text{and} \quad qs \neq 0,$$

where $pr, qs \in \mathbb{Z}$. Hence $ab \in \mathbb{Q}$.

By the contrapositive,

$$\forall a, b \in \mathbb{R}, (ab \notin \mathbb{Q}) \implies (a \notin \mathbb{Q} \vee b \notin \mathbb{Q}).$$

Q.E.D.

Definition 4.2.7 (Elimination Method)

Let A, B, C be statement variables.

$$(A \Rightarrow (B \vee C)) \equiv ((A \wedge \neg B) \Rightarrow C)$$

This is useful when the conclusion is a disjunction $B \vee C$: instead of proving the disjunction directly, we assume one disjunct fails ($\neg B$) and show the other (C) must then hold.

Since $B \vee C \equiv C \vee B$, we may instead eliminate C and prove

$$(A \wedge \neg C) \Rightarrow B.$$

Example 4.2.8

Here is an example:

Proposition 4.2.9

Prove that for all $x \in \mathbb{R}$, if $|2x - 6| = 4$, then $x > 3$ or $x = 1$.

Proof

The negation of $x > 3$ is $x \leq 3$, so the elimination form is

$$\forall x \in \mathbb{R}, [(|2x - 6| = 4) \wedge (x \leq 3)] \Rightarrow x = 1.$$

Let x be an arbitrary real number. Assume $|2x - 6| = 4$ and $x \leq 3$, and prove $x = 1$.

Since $|2x - 6| = 4$, thus

$$2x - 6 = \pm 4.$$

So $2x = 10$ or $2x = 2$, meaning

$$x = 5 \quad \text{or} \quad x = 1.$$

Since $x \leq 3$, the value $x = 5$ is ruled out, so

$$x = 1.$$

Q.E.D.

Proposition 4.2.10

$$\forall a, b, c \in \mathbb{Z}, (\forall x \in \mathbb{Z}, a \mid (bx + c)) \Rightarrow (a \mid (c - 2b)).$$

Remark (Proof idea)

The hypothesis holds for *every* integer x , so we may instantiate it at any x we like. Choosing $x = -2$ makes $bx + c$ collapse to $c - 2b$, which is exactly the divisibility claim we want — no other reasoning is needed once $x = -2$ is substituted.

Proof

Let a, b, c be arbitrary integers. Assume for all $x \in \mathbb{Z}$, $a \mid (bx + c)$.

Since this holds for every integer x , it holds in particular when $x = -2$, so we get

$$a \mid (b(-2) + c),$$

that is

$$a \mid (c - 2b).$$

Q.E.D.

Proposition 4.2.11 (Divisibility of Integer Combinations (DIC))

Let $a, b, c \in \mathbb{Z}$. If $a \mid b$ and $a \mid c$, then for all $x, y \in \mathbb{Z}$,

$$a \mid (bx + cy).$$

Remark (Proof idea)

Divisibility means b and c are each an integer multiple of a . Any integer combination $bx + cy$ is then a sum of multiples of a , hence itself a multiple of a — this is exactly what licenses factoring a out below.

Proof

Let $a, b, c \in \mathbb{Z}$. Assume $a \mid b$ and $a \mid c$. Then there exist integers k_1, k_2 such that

$$b = ak_1 \quad \text{and} \quad c = ak_2.$$

Let $x, y \in \mathbb{Z}$ be arbitrary. Then

$$bx + cy = ak_1x + ak_2y = a(k_1x + k_2y).$$

Since $k_1x + k_2y \in \mathbb{Z}$, this shows $a \mid (bx + cy)$.

Q.E.D.

Proposition 4.2.12 (Converse of DIC)

Let $a, b, c \in \mathbb{Z}$. If for all $x, y \in \mathbb{Z}$, $a \mid (bx + cy)$, then $a \mid b$ and $a \mid c$.

Remark (Proof idea)

The hypothesis holds for *every* choice of x, y , so we are free to pick whichever values isolate b and c on their own: $(x, y) = (1, 0)$ leaves $bx + cy = b$, and $(x, y) = (0, 1)$ leaves $bx + cy = c$.

Proof

Let a, b, c be arbitrary integers. Assume that for all $x, y \in \mathbb{Z}$, $a \mid (bx + cy)$.

Taking $x = 1, y = 0$, we get

$$a \mid (b \cdot 1 + c \cdot 0), \quad \text{that is,} \quad a \mid b.$$

Taking $x = 0, y = 1$, we get

$$a \mid (b \cdot 0 + c \cdot 1), \quad \text{that is,} \quad a \mid c.$$

Therefore $a \mid b$ and $a \mid c$.

Q.E.D.

§4.3 Proof By Contradiction

Definition 4.3.1

Given a statement A , the logical expression $A \wedge (\neg A)$ is called a contradiction.

A contradiction is false in every row of its truth table. To prove a statement S by contradiction, assume $\neg S$ and derive a contradiction; since a true assumption cannot lead to a false statement, $\neg S$ must be false, so S is true. For an implication $A \implies B$, we assume A together with $\neg B$.

Proposition 4.3.2

$\forall n \in \mathbb{Z}$, if n^2 is even, then n is even.

Proof

Let n be an arbitrary integer.

Assume n^2 is even.

Suppose, for contradiction, that n is odd.

Then,

$$(\exists k \in \mathbb{Z}) n = 2k + 1.$$

It implies

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

but $2k^2 + 2k \in \mathbb{Z}$ (since $k \in \mathbb{Z}$). So we have actually shown that n^2 is odd.

This contradicts the assumption that n^2 is even.

Thus, the assumption that n is odd must be false. Thus, n is even.

Q.E.D.

Exercises

- Prove there is no greatest integer.
- Prove $\forall a, b \in \mathbb{Z}, a \geq 2 \implies (a \nmid b) \vee (a \nmid (b + 1))$.
- Prove $\sqrt{2}$ is irrational.

§4.4 Proving If and Only If Statements

The statement $A \iff B$ means $(A \implies B) \wedge (B \implies A)$. Prove both directions to establish the equivalence.

Example 4.4.1

Prove that for all non-negative $x, y \in \mathbb{R}$,

$$\frac{x+y}{2} = \sqrt{xy} \iff x = y.$$

Remark (Proof idea)

If $x = y$, evaluate both sides. In the other direction, square the equality and rearrange it into $(x - y)^2 = 0$.

Proof

Let $x, y \in \mathbb{R}$ be arbitrary and non-negative.

($\Leftarrow$) Suppose $x = y$. Then

$$\frac{x+y}{2} = y = \sqrt{y^2} = \sqrt{xy},$$

where $\sqrt{y^2} = y$ because $y \geq 0$.

($\Rightarrow$) Suppose $\frac{x+y}{2} = \sqrt{xy}$. Squaring both sides gives

$$x^2 + 2xy + y^2 = 4xy,$$

so $(x - y)^2 = 0$. Hence $x = y$.

Q.E.D.

§4.5 Proving Uniqueness

To prove that there is a unique $x \in S$ satisfying $P(x)$, show both that such an x exists and that any two elements satisfying P are equal. Proving only the second part does not establish existence.

Proposition 4.5.1

For all $a, b \in \mathbb{Z}$, if $a \neq 0$ and $a \mid b$, then there is a unique integer k such that $b = ka$.

Remark (Proof idea)

Divisibility gives existence directly. For uniqueness, compare two possible multiples of the same nonzero integer a .

Proof

Let $a, b \in \mathbb{Z}$ be arbitrary. Assume $a \neq 0$ and $a \mid b$.

Existence. By the definition of divisibility, there is some $k \in \mathbb{Z}$ such that $b = ka$.

Uniqueness. Suppose $k, m \in \mathbb{Z}$ both satisfy the equation. Then $ka = b = ma$. Since $a \neq 0$, dividing by a gives $k = m$.

Q.E.D.

Remark

The condition $a \neq 0$ is needed for uniqueness. If $a = b = 0$, then $0 \mid 0$, but every integer k satisfies $0 = k \cdot 0$.

Exercises

- (i) Use a proof by contradiction to show that two non-parallel lines have a unique intersection point.

§5 Sets

§5.1 Basic Set Notation and Operations

Definition 5.1.1

A *set* is a collection of distinct elements. The *empty set* $\emptyset$ has no elements. For a finite set S , its *cardinality* $|S|$ is its number of elements.

We can describe a set by listing its elements, selecting elements with a property, or generating values:

- Listing: $\{2, 3, 4, 6\}$.
- Selection: $\{n \in \mathbb{Z} : 0 < n < 10 \text{ and } n \text{ is even}\} = \{2, 4, 6, 8\}$.
- Generation: $\{2k : k \in \{1, 2, 3, 4\}\} = \{2, 4, 6, 8\}$.

Example 5.1.2

Order and repetition do not change a set: $\{1, 2, 2\} = \{2, 1\}$. Also, $|\emptyset| = 0$, while $|\{\emptyset\}| = 1$: the latter set contains the empty set as its one element.

§5.1.1 Set-Builder Notation

Definition 5.1.3

The *selection form* $\{x \in U : P(x)\}$ contains exactly the elements of U for which $P(x)$ is true. The *generation form* $\{f(x) : x \in U\}$ contains the outputs of f , which need not be the inputs.

Example 5.1.4

$\{x^2 : x \in \{-2, -1, 0, 1, 2\}\} = \{0, 1, 4\}$: different inputs may give the same output, which appears only once.

We can combine the forms: $\{f(x) : x \in U, P(x)\}$ first selects the inputs satisfying P , then forms their outputs. For example,

$$\{3a : a \in \mathbb{Z}, 2 \mid a\} = \{6k : k \in \mathbb{Z}\}.$$

§5.1.2 Intervals

For real numbers $a < b$, square brackets include an endpoint and round brackets exclude it:

$$\begin{aligned} [a, b] &= \{x \in \mathbb{R} : a \leq x \leq b\}, & (a, b) &= \{x \in \mathbb{R} : a < x < b\}, \\ [a, b) &= \{x \in \mathbb{R} : a \leq x < b\}, & (a, b] &= \{x \in \mathbb{R} : a < x \leq b\}. \end{aligned}$$

§5.1.3 Set Operations

Definition 5.1.5

The *union* $S \cup T$ collects elements in either set; the *intersection* $S \cap T$ keeps elements in both. The *difference* $S - T$ keeps elements in S but not T :

$$S \cup T = \{x : x \in S \text{ or } x \in T\}, \quad S \cap T = \{x : x \in S \text{ and } x \in T\},$$

$$S - T = \{x : x \in S \text{ and } x \notin T\}.$$

When S is a subset of a universal set U , its *complement* is $\bar{S} = U - S$.

Example 5.1.6

Let $A = [-1, 2)$ and $B = (0, 4]$. Then

$$A \cup B = [-1, 4], \quad A \cap B = (0, 2),$$

$$A - B = [-1, 0], \quad B - A = [2, 4].$$

The point 2 belongs to B but not A , so it is excluded from the intersection and included in $B - A$.

Some useful identities, with complements taken relative to U , are

$$\begin{aligned} S \cup \emptyset &= S, & S \cap \emptyset &= \emptyset, \\ S \cup U &= U, & S \cap U &= S, \\ S \cup S &= S, & S \cap S &= S, \\ S \cup \bar{S} &= U, & S \cap \bar{S} &= \emptyset. \end{aligned}$$

§5.1.4 Subset**Definition 5.1.7 (Disjoint)**

Two sets S and T are said to be disjoint when

$$S \cap T = \emptyset$$

Definition 5.1.8 (Subset)

A set S is called a subset of a set T when every element of S belongs to T .

$$S \subseteq T.$$

Definition 5.1.9 (Proper Subset)

A set S is called a proper subset of a set T when S is a subset of T and there exists an element in T which does not belong to S . We write

$$S \subsetneq T$$

Remark

When we have $S \subseteq T$: we also say T is superset of S .

When we have $S \not\subseteq T$: we say T is a proper superset of S .

Example 5.1.10 – $S \subseteq S$ for all sets

- $\emptyset \subseteq S$ for all sets.
- $\mathbb{N} \not\subseteq \mathbb{Z}$
- $\mathbb{Z} \not\subseteq \mathbb{N}$
-

To prove that $S \subseteq T$, prove the universally quantified implication

$$\forall x \in \mathcal{U}, (x \in S) \implies (x \in T)$$

Example 5.1.11

Let $A = \{n \in \mathbb{N} : 4 \mid (n - 3)\}$ and $B = \{2k + 1 : k \in \mathbb{Z}\}$.

- (i) Prove that $A \subseteq B$
- (ii) Prove that $A \subseteq B$

Proof

For (i)

Let $n \in \mathbb{N}$. Assume $n \in A$.

Then by the definition of the set, $4 \mid (n - 3)$, which means $\exists k \in \mathbb{Z}$ such that

$$\begin{aligned} n - 3 &= 4k \implies n = 4k + 3 = 4k + 2 + 1 \\ n &= 2(2k + 1) + 1. \end{aligned}$$

Since $2k + 1 \in \mathbb{Z}$, $n \in B$.

Therefore, $A \subseteq B$.

For (ii) If $n = 1$, $1 \in B$.

$$4 \nmid 1 - 3 \implies 1 \notin A. \therefore A \subsetneq B$$

Q.E.D.

In order to prove $S = T$, we need to prove

$$S \subseteq T \quad T \subseteq S.$$

Example 5.1.12

Let $S = \{1, -1, 0\}$ and $T = \{x \in \mathbb{R} : x^3 = x\}$. Prove that $S = T$.

Proof

- $S \subseteq T$

Let $x \in S$. Then $n = 1, -1, 0$.

$$x = 1 \implies 1^3 = 1 \implies x \in T$$

$$x = 0 \implies 0^3 = 0 \implies x \in T$$

$$x = -1 \implies (-1)^3 = -1 \implies x \in T$$

– $T \subseteq S$

Let $x \in T$. Then $x^3 = x$. So,

$$x^3 - x = 0$$

$$x(x^2 - 1) = 0$$

$$x(x - 1)(x + 1) = 0$$

Therefore, $x = 0, 1, -1$.

that means $x \in S$. Therefore, $T \subseteq S$.

Therefore $S = T$.

Q.E.D.

Example 5.1.13

Prove that $S = T$ if and only if $S \cap T = S \cup T$.

endequation*

Proof

($\implies$) Suppose $S = T$.

First we prove $(S \cap T) \subseteq S \cup T$.

Let $x \in S \cap T$. It implies $x \in S$ and $x \in T$.

It also implies $x \in S \vee x \in T \implies x \in S \cup T$

Now, we need to prove $S \cup T \subseteq S \cap T$

$$x \in S \cup T \implies x \in S \vee x \in T.$$

Since $S = T$,

$$\implies x \in S \wedge x \in T \implies S \cap T.$$

Alternative:

$$S \cap T = S \cap S = S \cup S = S \cup T.$$

($\impliedby$) Suppose $S \cup T = S \cap T$.

First $S \subseteq T$:

$$x \in S \implies x \in S \cup T \implies x \in S \cap T \implies x \in T$$

Kiro, you do the second way.

Q.E.D.

§6 Mathematical Induction

§6.1 Principle of Mathematical Induction

Axiom (The Principle of Mathematical Induction)

Let $P(n)$ be an open sentence that depends on $n \in \mathbb{N}$. If statements (1) and (2) are true

(1) $P(1)$

(2) $P(K) \implies P(K + 1), \forall n \in \mathbb{N}$

Then, $P(n), \forall n \in \mathbb{N}$.

Prove by Mathematical Induction

To prove the universally quantified statements " $\forall n \in \mathbb{N}, P(n)$ "

– Prove $P(1)$: Base Case.

– Prove the Universally quantified implication. "For all $k \in \mathbb{N}$, If $P(k)$, then $P(k+1)$ "

Example 6.1.1

Prove for all $n \in \mathbb{N}$

$$\sum_{i=1}^n i(i+1) = \frac{1}{3}n \cdot (n+1) \cdot (n+2)$$

Proof

Base case: $P(1)$ is true since

$$\sum_{i=1}^1 i \cdot (i+1) = 1 \cdot 2 = 2 = \frac{1}{3} \cdot 1 \cdot (1+1) \cdot (1+2)$$

Inductive Step

Let k be an arbitrary natural number.

Assume $P(k)$

That is, assume

$$\sum_{i=1}^k i(i+1) = \frac{1}{3}k(k+1)(k+2)$$

Prove $P(k+1)$:

$$\begin{aligned} \sum_{i=1}^{k+1} i(i+1) &= \sum_{i=1}^k i(i+1) + (k+1)(k+2) \\ &= \frac{1}{3}k(k+1)(k+2) + (k+1)(k+2) \\ &= (k+1)(k+2)(k+1) \cdot \frac{1}{3} \\ &= \frac{1}{3}(k+1)(k+2)(k+3) \end{aligned}$$

Therefore, $P(k + 1)$ is true.
Thus, $P(n)$ is true $\forall n \in \mathbb{N}$ by POMI.

Q.E.D.

Example 6.1.2

Prove $n! > 2^n$ for all integer $n \geq 4$.

Proof

By Induction on n

Base Case: Show that stmt is true for $n = 4$.

$$\text{LS} = 4! = 24$$

$$\text{RS} = 2^4 = 16$$

Since $24 > 16$, the stmt is true for $n = 4$.

Inductive Step: Let k be an arbitrary natural number with $k \geq 4$.

Assume the stmt is true for $n = k$.

That is, assume $k! > 2^k$.

Now prove the stmt is true for $k + 1$.

That is, prove $(k + 1) > 2^{k+1}$

We know

$$k! > 2^k$$

by I. H

$$(k + 1) k! > (k + 1) 2^k$$

$$(k + 1)! > (k + 1) 2^k$$

Since $k \geq 4$, $k + 1 \geq 5 \geq 2$, so $k + 1 \geq 2$.

So

$$(k + 1)! > (k + 1) 2^k \geq 2 \cdot 2^k = 2^{k+1}$$

Q.E.D.

§6.2 Principle of Strong Induction

Why Mathematical Induction is not enough?

Example 6.2.1

A sequence $\{x_n\}$ is defined by

$$x_1 = 4 \quad x_2 = 68 \quad x_n = 2x_{n-1} + 15x_{n-2} \quad \forall n \geq 3$$

Prove $\underbrace{x_n}_{\text{closed form}} = 2(-3)^n + 10 \cdot 5^{n-1}, \forall n \geq 1.$

Remark

This is an example of a sequence defined *recursively* or a recursive relation.

Proof

Let $P(n)$ be $x_n = 2(-3)^n + 10 \cdot 5^{n-1}$

Base case: Prove $P(1)$ is true. We have

$$x_1 = 4$$

by definition. also

$$2 \times (3)^1 + 10 \cdot 5^{1-1} = 2(-3) + 10 = -6 + 10 = 4 = x_1$$

So $P(1)$ is true.

Inductive Step Let k be an arbitrary natural number.

Assume $P(k)$ is true.

That is

$$x_k = 2(-3)^k + 10 \cdot 5^{k-1}.$$

Prove $P(k+1)$. That is

$$x_{k+1} = 2(-3)^{k+1} + 10 \cdot 5^k.$$

Q.E.D.

We cannot continue with this. Since we are unable to prove that

$$P(k) \implies P(k+1)$$

since we only know $P(k)$ is true. If we want or prove, we also need to know that $P(k-1)$ is true.

Definition 6.2.2 (Principle of Strong Induction)

Let $P(n)$ be an open sentence that depends on $n \in \mathbb{N}$.

If statement 1 and 2 are both true

- (i) $P(1)$
- (ii) For all $k \in \mathbb{N}$, if

$$P(1) \wedge P(2) \wedge \cdots \wedge P(k)$$

is true, then $P(k+1)$ is true.

then $P(k), \forall k \in \mathbb{N}$.

Prove by Strong Induction

Prove $\forall n \in \mathbb{N}, n \geq b, P(n)$

- Prove $P(b) \wedge P(b+1) \wedge \cdots \wedge P(B), \forall B \geq b$.
- Inductive step: Prove for all integers $k \geq B$.

Go back to the bloody example.

Example 6.2.3

A sequence $\{x_n\}$ is defined by

$$x_1 = 4 \quad x_2 = 68 \quad x_n = 2x_{n-1} + 15x_{n-2} \quad \forall n \geq 3$$

Prove $\underbrace{x_n}_{\text{closed form}} = 2(-3)^n + 10 \cdot 5^{n-1}, \forall n \geq 1.$

Proof

Proof by induction on n .

Base Case: Prove $P(1)$ By definition,

$$\begin{aligned} 2 \times (-3)^1 + 10 \cdot 5^{1-1} &= 2(-3) + 10 = -6 + 10 = 4 = x_1 \\ 2 \times (3)^2 + 10 \cdot 5^{2-1} &= 18 + 50 = 68 = x_2 \end{aligned}$$

Therefore, $P(1)$ is true.

Inductive step : Let k be an arbitrary integer with $k \geq 2$

Assume $P(i)$ is true for all integer i with $1 \leq i \leq k$.

That is assume

$$x_i = 2(-3)^i + 10 \cdot 5^{i-1}, \forall i \in \{1, 2, 3, \dots, k\}$$

Prove $P(k+1)$ is true. That is

$$x_{k+1} = 2(-3)^{k+1} + 10 \cdot 5^k.$$

By define

$$\begin{aligned} x_{k+1} &= 2x_k + 15x_{k-1} \\ &= 2[2(-3)^k + 10 \cdot 5^{k-1}] + 15 \cdot 2[2(-3)^{k-1} + 10 \cdot 5^{k-2}] \\ &= \text{Chatgpt, help me expand it.} \\ &= 2(-3)^{k+1} + 10 \cdot 5^k \end{aligned}$$

for $k \geq 2$.

$P(k+1)$ is true. So this $P(n)$ is true for all n in $\mathbb{N}$.

Q.E.D.

Example 6.2.4

Suppose $x_1 = 3, x_2 = 5$ and $x_n = 3x_{n-1} + 2x_{n-2}, \forall n \geq 3$.

Prove $x_n < 4^n$ for all n in $\mathbb{N}$.

Example 6.2.5

Suppose you have a $m \times n$ rectangular chocolate bar with mn unit square.

Let $N = mn$. Prove $N - 1$ breaks required to break the choc bar into unit square.